

## HS 528: Information Technology and Development

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| Course Code   | HS 528                                                           |
| Course Name   | Information Technology and Development                           |
| L-T-P-C       | 3-0-0-3                                                          |
| Prerequisites | None                                                             |
| Intended for  | Post-graduate students; 3rd and 4th Year B. Tech. students       |
| Distribution  | Core course for M.A. in Development Studies, Elective for others |
| Approval      | 23rd BoA (Apr 2018); 16th Senate (Apr 2018)                      |
| Revised       | 50th BoA (Jul 2023) - revised                                    |

### Course Contents

- **Introduction to technology and development** Definitions of technology, Systemic views of Technology-Society relationships, Development paradigms, Social and commercial mechanisms for linking technology to developmental needs; (Certain example cases could include: a case study on same language subtitling (SLS) for improving reading ability in India; a case study on hole-in-the-wall project in India, Bhutan, Cambodia, and Africa and how it encourages children to learn via entertainment; self-organized learning and mediated environments in India and Oscar-winning movie Slumdog Millionaire; failure of little intelligent community based ICT in Central America.) [8 Lectures]
- **Overview of ICT for development** Development agenda and positioning of ICT for development; information and knowledge society; technological interests and social change; real-world applications of ICT in development; the role of technology in reducing poverty (e.g., computer aided learning and Mille.org); the role of ICT in overcoming healthcare challenges (e.g., telemedicine, information gathering, and low-cost point of care testing medical devices- translating health care from bench to bedside); the role of technology in causing agricultural innovation (e.g., telecentres and digital Green like the green-wifi.org and the e-sagu project); role of ICT in surveillance and good governance, communications, infrastructure, and user-interface design. [12 Lectures]
- **Participatory methods in technology and development** Discussion on participatory methods (e.g., documentary Water of Ayol (Sandra Nichols)); different principles and modes of participation; contributors to ICT projects failures; role and issues in community-based participatory research (e.g., case of the African Digital Renaissance). [5 Lectures]
- **Networking, access, mobile technologies, and development** Access to hardware, software, information and services; LTE and Internet backbone; role of TCP/IP and other protocols; (Certain example cases could include the case of the last 10km in Rwanda; case of microwave in rural Alaska; role of cellular voice and data; case of M-PESA in India and Kenya.) [5 Lectures]
- **Cyber-security and development** Intrusion-detection systems; deception; block chain technology; Cloud access security brokers (CASB); Endpoint detection and response (EDR); Non-signature approaches for endpoint prevention; Remote browser. [3 Lectures]
- **User interface design, cloud computing, HCI and development** Role of cloud computing in promoting development; configuring Wikispaces and a website; role of HCI in development since its emergence in 1980s; symbiotic relationship between humans and computers (example, the case of spoken web: using voice as an accessibility tool for disadvantaged people in developing countries); grand challenges in HCI for development. [4 Lectures]

- **New devices, sensor networks in development** Role of sensor and IoT revolution and how it aids development (examples of IoT for farming and natural disasters); notion of wearable computing devices and its impact on sustainable development; wireless sensor networks and their role in development; influence of IoT on different policy areas. [5 Lectures]

### Text Books

1. Heeks, R. (2017). Information and Communication Technology for Development (ICT4D). Routledge, London, UK.
2. Seth, K. (2016). Computers, Internet And New Technology Laws-A Comprehensive Reference Work With Special Focus On Developments In India. Lexis Nexis

### References

1. Barrat, J., Our final invention: Artificial intelligence and the end of the human era. Macmillan, 2013.
2. Cooper, C., Technology and development in the Third industrial revolution. Routledge, 2017.
3. Cornwall, A., & Jewkes, R., What is participatory research?. Social science & medicine, 41(12), 1667-1676, 1995.
4. Dodson, L. L., Sterling, S., & Bennett, J. K., Considering failure: eight years of ITID research. In Proceedings of the fifth international conference on information and communication technologies and development (pp. 56-64). ACM, 2012.
5. Garrity, J., Harnessing the Internet of Things for Global Development. SSRN Electronic Journal. doi:10.2139/ssrn.2588129, 2015.
6. Ho, M. R., Smyth, T. N., Kam, M., & Dearden, A., Human-computer interaction for development: The past, present, and future. Information Technologies & International Development, 5(4), pp-1, 2009.
7. Kendall, J., & Voorhies, R., The mobile-finance revolution: How cell phones can spur development. Foreign Affairs, 93(2), 9-13, 2014.
8. Kumar, A., & Agarwal, S. K., Spoken web: using voice as an accessibility tool for disadvantaged people in developing regions. ACM SIGACCESS Accessibility and Computing, (104), 3-11, 2012.
9. Mandal, G. C., Social Impact of Banking, Technology and Development In Village India. Rajat Publications, 2012.
10. Mas, I., & Radcliffe, D. (2010). Mobile payments go viral: M-PESA in Kenya. The Capco Institute Journal of Financial Transformation, 169-182, 2010.
11. Rathakri & Thomas, M., Participatory Technology Development: A Technique for Indigenous Technical Knowledge Refinement. Scientific Publisher, 2012.
12. Pannu, P., & Tomar, Y. A., Communication Technology for Development. I K International Publishing House Pvt. Ltd, 2012.
13. Raina, R. S., Science, Technology and Development in India: Encountering Values. Orient Longman, 2015.
14. Sankar, S., The rise of human-computer cooperation. In TED Talk Lesson, TED Conferences, junho de, 2012.
15. Smith, M. L., Spence, R., & Rashid, A. T., Mobile phones and expanding human capabilities. Information Technologies & International Development, 7(3), pp-77, 2011.
16. Toyama, K., On turbocharged, heat-seeking, robotic fishing poles. Communications of the ACM, 54(12), 29-31, 2011.
17. Unwin, T., ICT4D: Information and Communication Technologies for Development. Cambridge University Press, Cambridge, 2008.